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Journal of Educational Computing Research
2026, Vol. 64(2) 311–343
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Asking, Playing, Learning: Investigating Large Language Model-Based Scaffolding in Digital Game-Based Learning for Elementary Artificial Intelligence Education

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Abstract

Digital game-based learning (DGBL) has demonstrated notable effectiveness in general artificial intelligence (AI) literacy education, but it often falls short in addressing personalized learning needs. Traditional game-based scaffolding lacks real-time feedback, thereby limiting the students' deep cognitive engagement. To address these limitations, this study introduces a large language model (LLM)-based scaffolding designed to create an engaging and interactive AI learning environment. Using a quasi-experimental design with fifth-grade students, we compared an experimental group using LLM-based scaffolding (LLM-DGBL) with a control group using traditional scaffolding (TS-DGBL). The results showed that the experimental group outperformed the control group in both learning achievement and cognitive load reduction. Behavioral analysis showed that students in the experimental group engaged in more diverse and deeper self-regulated learning patterns. Moreover, interaction analysis revealed a key pattern: students with a higher cognitive load sought support from the

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LLM-based scaffolding more frequently than their peers with a lower cognitive load. This study presents a novel method for elementary AI literacy education. It also provides a valuable reference for applying generative AI in educational practice to support the development of adaptive learning tools tailored for young learners.

Keywords
artificial intelligence education, digital game-based learning, large language model, large language model-based scaffolding

Introduction

Artificial Intelligence (AI) literacy refers to an individual's integrated capacity to leverage AI technologies for solving real-world problems (Wang et al., 2023). As AI penetrates ever more deeply into diverse domains, acquiring foundational AI knowledge and cultivating AI-driven thinking and practical skills have become core competencies in the digital era (Davenport & Ronanki, 2018). However, many real-world challenges in AI curriculum design and implementation are seriously impeding the effective enhancement of AI literacy. In particular, most AI courses are confined to higher education, while adolescence—a key stage for developing AI interest—remains mostly neglected (Su et al., 2024). This is particularly critical for elementary education, which represents a foundational period for cognitive development. However, young learners often struggle with the abstract nature of AI concepts. Therefore, this study addresses their specific learning needs and cognitive characteristics by employing a DGBL approach, which transforms abstract knowledge into engaging, concrete tasks suitable for their developmental readiness.

Moreover, traditional instructional approaches tend to be uniform and lack the diversity and interactivity needed to support students' intuitive grasp and application of complex AI concepts, resulting in skill development that falls behind industry demands (Casal-Otero et al., 2023). To address these challenges, digital game-based learning (DGBL) has been introduced into AI learning. Empirical evidence shows that this approach enhances the effectiveness of AI courses and improves both student performance and motivation (Chen et al., 2023; Kahyaoğlu Erdoğmuş & Kurt, 2024; Wang & Lester, 2023).

DGBL is a pedagogical model defined by the coming together of serious learning and interactive entertainment (Prensky, 2003). This approach facilitates the understanding of new concepts and has emerged as a leading pedagogical model in contemporary education (Yang et al., 2018). Empirical studies demonstrate its effectiveness across disciplines—improving mathematics performance (Hussein et al., 2022), strengthening vocabulary acquisition (Zou et al., 2021), and supporting scientific inquiry (Yeoh et al., 2024)—with consistent improvements in achievement, satisfaction, and motivation (Amzalag et al., 2024; Chang et al., 2020; Hung et al., 2015). Furthermore, the immersive contexts created by DGBL have been demonstrated to enhance students' flow experiences, effectively reduce cognitive load (Pan et al., 2025;

Wang et al., 2025), improve learning achievements (Chen & Hsu, 2020), and positively influence learning behaviors (Kahyaoğlu Erdoğmuş & Kurt, 2024).

Despite the great potential of DGBL, learners often face the challenge of diminished learning outcomes due to the high level of autonomy (Wang, 2020). This is largely attributed to complex game tasks, limited experience, and insufficient prior knowledge. To address this issue, scholars integrate scaffolding—an instructional method that provides tiered prompts and real-time feedback—into DGBL (Chen et al., 2023; Yeoh et al., 2024). By providing tiered prompts and real-time feedback, scaffolding can prevent students from losing direction due to excessive autonomy, regulate task difficulty, and reduce cognitive overload, thereby supporting the coordinated development of cognitive, affective, and metacognitive skills (Cai et al., 2022; Chang & Yang, 2023).

The introduction of scaffolding is beneficial in enhancing the learning outcomes in DGBL. However, traditional scaffolding often relies on manual design, employs rigid rules, and offers limited depth of response (Ding & Yu, 2024). Therefore, how to provide in-depth response and personalized guidance for learners within DGBL has become a critical problem (Ku et al., 2016; Yang & Chen, 2020). Recent studies indicate that Large Language Models (LLMs) can leverage their advanced language comprehension and knowledge to provide personalized support tailored to learners' real-time needs (Kumar et al., 2024; Mannekote et al., 2024). A promising application of this capability is LLM-based scaffolding. It uses LLMs to provide dynamic, context-aware, and personalized assistance—such as hints, feedback, and explanations—in real time (Wang et al., 2025). Empirical evidence for this approach shows that such systems can mitigate the decline in learning motivation associated with a lack of individualized attention. Furthermore, they have been shown to enhance learning outcomes (Wang et al., 2024).

However, while the potential of LLM-based scaffolding is widely acknowledged, there is a lack of empirical research that systematically investigates its effects within DGBL environments compared to traditional scaffolding. Specifically, it remains unclear how LLM-based dynamic support impacts key learning variables such as learning achievement, flow experience, and cognitive load in an integrated manner.

This study proposes an LLM-based scaffolding supported DGBL (LLM-DGBL) framework that methodologically integrates dynamic scaffolding mechanisms powered by large language models into DGBL environments. This integration enhances AI courses by making them more engaging, interactive, immersive, and appropriately challenging. Leveraging the advanced conversational and natural language processing capabilities of LLMs, LLM-based scaffolding provides timely feedback to deepen the students' understanding of core concepts of AI courses. We implemented this approach in an AI literacy course for elementary school students and conducted an experimental evaluation to assess its effectiveness. To systematically evaluate the impact of this novel approach, the present study is centered on the following research questions:

(1) Compared with traditional scaffolding supported DGBL (TS-DGBL), does LLM-DGBL lead to greater improvements in students' learning achievement?
(2) Compared with TS-DGBL, does LLM-DGBL lead to greater improvements in students' flow experience?

(3) Compared with TS-DGBL, does LLM-DGBL lead to greater reduction in students' cognitive load?
(4) Compared with TS-DGBL, does LLM-DGBL lead to greater promotion of students' active and sustained learning behavior?
(5) Do students with higher cognitive load interact more frequently with the LLM-based scaffolding compared to those with lower cognitive load?

To address these questions, this paper is structured as follows. We begin by reviewing the relevant literature on DGBL, game-based scaffolding, and LLMs. Following this, we provide a detailed description of the LLM-DGBL system developed for this study. Subsequently, the methodology is presented, outlining the experimental design and data analysis procedures. We then report the main findings concerning learning achievement, cognitive load, flow experience, and behavioral patterns. Finally, we discuss the implications of these findings, acknowledge the study's limitations, and conclude with directions for future research.

Literature Review

Educational Applications of DGBL

DGBL integrates digital game contexts and mechanics with educational objectives to create a game-based learning environment that combines both entertainment and education (Coleman & Money, 2020). Within this learning environment, students engage in interactive experiences and gain a sense of achievement and satisfaction through points and level-up rewards, while facilitating knowledge construction and skills development.

With the continuous advancement and application of AI technologies in various aspects of society, providing students with general education on AI has become increasingly important. Considering the limited cognitive capacity for complex AI concepts, introducing DGBL into AI education for elementary students holds significant pedagogical value (Alden & Tramonti, 2020; Zapušek & Rugelj, 2013). The contextual and challenging characteristics of DGBL not only help students gain a comprehensive understanding of cutting-edge AI technologies but also develop their computational and creative thinking skills (Dai et al., 2023; Yeoh et al., 2024).

Empirical studies have shown that DGBL can significantly enhance the learning motivation and programming proficiency of students (Kazimoglu et al., 2012). Hsu and Hsu (2024) integrated design thinking into a DGBL and observed that sixth-grade learners achieved notable gains in academic performance, computational thinking, and AI literacy. Yang et al. (2024) found that DGBL significantly enhances elementary students' self-regulated learning strategies (i.e., the methods learners use to actively manage their own learning process) and vocabulary levels. Notably, students with lower academic proficiency benefited more than their higher-performing peers, suggesting that DGBL can effectively narrow the gap between these groups.

Learners generally report that DGBL is more engaging than traditional classroom teaching (Chang et al., 2020; Hung et al., 2015). Moreover, empirical evidence indicates that DGBL positively impacts students' academic achievement (Amzalag et al., 2024), flow experience (Pan et al., 2025), cognitive load (Wang et al., 2025), and information literacy (Zou, Zhang, et al., 2021). Hung et al. (2015) incorporated a TPCs game into a mathematics course, successfully simulating students' interest in learning and significantly improving their flow experience, learning achievement, and satisfaction.

However, without adequate scaffolding, learners may encounter issues such as insufficient self-regulation, unclear learning goals, and increased anxiety due to excessive autonomy (Wang, 2020; Yang et al., 2018). This is particularly evident in DGBL, where learners facing high-difficulty tasks without timely feedback are likely to experience excessive cognitive load, which can dimmish learning outcomes (Davis et al., 2021). Therefore, to ensure students benefit from DGBL, it is essential to integrate scaffolding to guide learners' progress and promote effective self-regulated learning.

Game-Based Scaffolding Applications

Wood et al. (1976) proposed the concept of scaffolding to describe the transitional support provided by external sources when learners are unable to complete tasks independently. Since then, scaffolding has continued to receive significant attention in educational research (Chen & Hou, 2024; Kao et al., 2017; Saye & Brush, 2002). With the rise of computer-assisted learning and DGBL, scaffolding has evolved from its traditional interpersonal form to encompass diverse forms supported by advanced technology and tools (Chen et al., 2023).

In DGBL environments, various forms of scaffolding are employed to stimulate learning motivation, alleviate frustration, and integrate game mechanics with disciplinary content (Yeoh et al., 2024; Zhou & Lam, 2019). For instance, Chen et al. (2023) developed the mobile game *Guess My Rule*, embedding scaffolding through multiple representations—text, 2D images, and 3D models. Sun et al. (2011) incorporated three types of scaffolding—key-feature identification, frustration regulation, and operational demonstration—into the game *Professor Sudoku*. Experimental results indicated that these mechanisms significantly improved challenge levels and reduced frustration. Kao et al. (2017) implemented different types of scaffolding in the game *Crayon Physics Deluxe* and compared four conditions (demonstration, no-scaffolding, key-feature annotation, and non-game control), finding that the key-feature annotation group outperformed the demonstration group in conceptual understanding and the "sensitivity" dimension of creativity, whereas the demonstration group scored higher in the "flexibility" dimension.

Empirical evidence indicates that embedding scaffolding within DGBL can effectively support the learning process. However, traditional scaffolding typically depends on predefined rules and provides static feedback based on student performance indicators (such as response accuracy and task completion), making it difficult to adapt dynamically to individual differences (Saye & Brush, 2002). Consequently, the

integration of LLMs to create novel scaffolding has emerged as a promising research direction. For example, Chen and Chang (2024) introduced ChatGPT interactions into a science learning game, comparing two AI-assisted conditions (with and without examples) against a non-LLM control group. Their results demonstrated that students in the LLM-assisted groups exhibited stronger intrinsic motivation, lower cognitive load, and more efficient learning behaviors. Similarly, Chen and Hou (2024) found that LLMs can provide timely feedback and precise explanations, thereby enhancing learners' understanding and mastery.

Research on LLM-based scaffolding is still in an exploratory stage. There is a lack of systematic research on how to provide adaptive and targeted guidance in DGBL based on individual differences. Accordingly, this study aims to examine the interactional authenticity and educational effectiveness of LLM-based scaffolding in DGBL, to assess its potential to meet learners' personalized needs, and to provide both theoretical and empirical foundations for the optimized design of adaptive scaffolding.

LLMs

LLMs are intelligent models for natural language understanding and content generation. They are built on deep neural network architectures and pre-trained on large-scale corpora to learn language representation and generation (Naveed et al., 2024). Since Bengio et al. (2003) first introduced deep neural networks to language modeling, the representational capabilities of such models have progressively improved. The introduction of Transformer architecture (Vaswani et al., 2017) enabled more accurate modelling of syntactic and semantic relationships through large-scale pre-training. Building on this foundation, OpenAI's GPT series further expanded the capabilities of LLMs in handling complex tasks through few-shot learning and large-scale inference (Ouyang et al., 2022).

LLMs offer a series of core advantages. First, they can generate high-quality text based on prompts, supporting tasks such as writing, translation, and summarization (Tiwari et al., 2024). Second, their conversational interfaces provide timely feedback which enables personalized needs (Dehghani & Mashhadi, 2024). Third, leveraging vast pre-trained knowledge bases, LLMs can provide users with rich related information (Saif et al., 2024). These features highlight the great potential of LLMs in areas such as information retrieval, content creation, and decision support.

In educational settings, LLMs can provide personalized learning content based on learners' existing knowledge, learning preferences, and response pace, thereby enhancing both learning experience outcomes (Gao et al., 2024). For example, Guo et al. (2025) introduced LLMs into science and engineering practicum instruction, leading to significant improvements in middle school students' scientific knowledge, computational thinking, and problem-solving skills. Zhu et al. (2025) demonstrated the effectiveness of an LLM-based chatbot in increasing homework submission rates and academic performance in middle school homework. Yan et al. (2025) found that incorporating LLM into collaborative programming significantly reduced students' cognitive load while improving their computational thinking ability. In a randomized

controlled trial, Liu et al. (2024) showed that the LLM-supported Cognitive Academic Language Learning Model (CALLA-LLM) instructional model significantly outperformed the traditional CALLA in enhancing EFL writing performance, self-regulated learning strategy use, and writing motivation.

These findings indicate that the natural language interaction and real-time feedback capabilities of LLMs can provide flexible and personalized support for students, making them highly suitable for DGBL. However, empirical applications of LLMs in DGBL remain scarce. Therefore, the present study aims to develop an LLM-based scaffolding within a DGBL environment, enabling real-time learner diagnosis and dynamic feedback, and further evaluate its combined effects on learning achievement, flow experience, and cognitive load.

LLM-Based Scaffolding in DGBL

With the widespread adoption of DGBL in AI education (Wang et al., 2021), effective knowledge and skill development in game-based environments has become a key research focus (Dimitriadou & Lanitis, 2023). Integrating scaffoldings into DGBL has proven effective for enhancing the quality of elementary school AI curricula (Ait Baha et al., 2024; Chen & Chang, 2024). Accordingly, this study introduced LLM-based scaffolding to develop an educational game *AI World*.

DGBL engages students through game-based tasks, while real-time behavioral data collection enables support for self-regulated learning. Within this framework, LLM assists with knowledge-related queries, and the game's task design and feedback mechanisms further promote self-regulation. All interaction logs and performance data are systematically stored in a backend database, allowing teachers to visualize student trajectories and provide personalized guidance. The course was co-designed by the research team and frontline teachers to ensure strong alignment between the game and curriculum standards, thereby supporting the development of students' digital literacy and practical competencies.

DGBL Framework

DGBL typically comprises the following key elements: learner autonomy, contextualized narratives, clear educational objectives and rules, interactivity and challenge, immediate feedback, and a sense of achievement (Chen et al., 2023; Coleman & Money, 2020). In this study, a literature review and expert consultation were first conducted, providing both theoretical and practical foundations to the design of the testing tasks. Subsequently, an LLM-based scaffolding was developed for the DGBL. Within the DGBL framework, students can receive supportive prompts and responsive feedback to enhance learning outcomes when engaging in game tasks. Meanwhile, teachers can access real-time learning data via the system's backend. As shown in Figure 1, the system framework consists of three coordinated layers: a core processing layer, a data management layer, and a functional module layer. Teachers use the management interface to distribute questionnaires and game tasks, monitor student

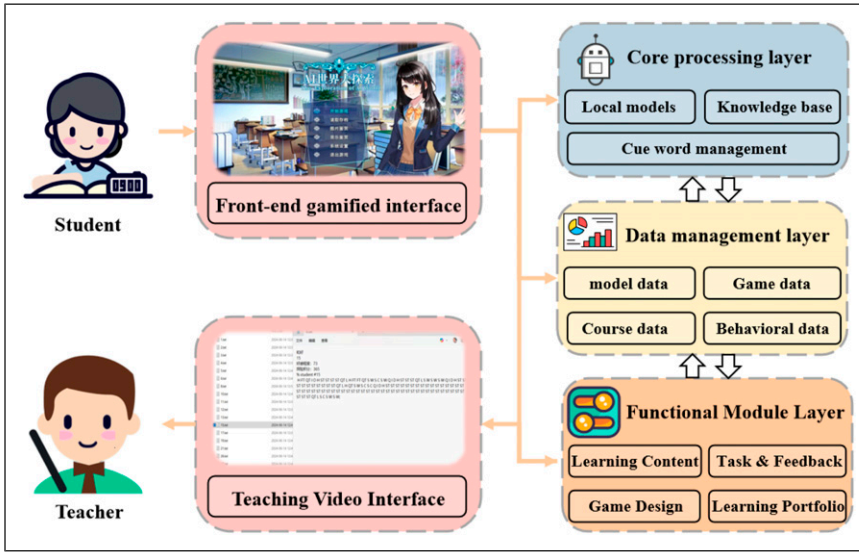


Figure 1. DGBL Framework

progress, and collect learning data, while students engage the DGBL through an immersive gamified interface.

Description of LLM-Based Scaffolding

At the start of the game, the system employs animation and voice-over narration to present the background story, game rules, and final objectives of *the AI world*, helping students understand their role as explorers tasked with investigating the field of AI and bringing the acquired knowledge back to the human world.

Once inside the game, students are required to complete a series of levels within a virtual environment, such as "What Is AI?", "Thinking Like a Human", and "Understanding Human Language." Students proceed by studying designated learning materials, watching micro-lectures, and solving clue-based tasks to unlock the next stages. To ensure the integrity of knowledge acquisition, the game allows only speed adjustment for video playback and does not permit skipping.

In the experimental group, students can summon and interact with a virtual intelligent avatar named "Mini-Xiaowei" (see Figure 2). Powered by an LLM, Mini-Xiaowei can generate personalized, real-time responses to students' questions and encourage interactions through varied facial expressions and tone of voice. Moreover, the rich textual outputs from the LLM can be converted into speech using the iFlytek API, allowing students to switch between auditory and visual modes based on individual preferences. In contrast, the control group can only access a static question bank with pre-defined information.

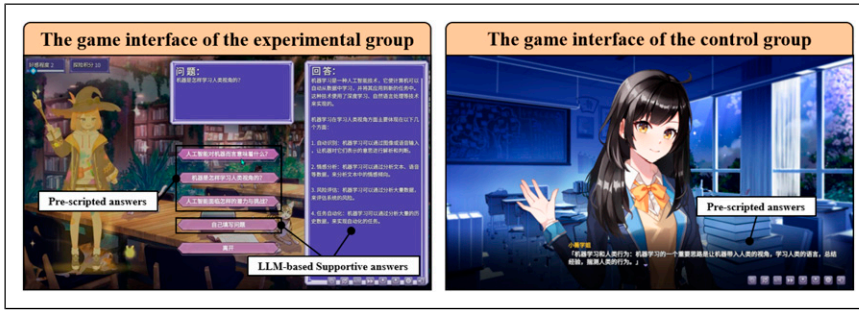


Figure 2. Comparison of DGBL Interfaces Between Experimental Group (LLM-Based DGBL, Left) and Control Group (TS-DGBL, Right)

When students reach a checkpoint, the system prompts them to watch a micro-lecture or interact with Mini-Xiaowei and complete a set of testing questions. As shown in Figure 3, the LLM-based scaffolding enables a more dynamic and engaging learning experience in experimental group. For instance, if a student incorrectly answers the question "What part of the human brain is most similar to the neural networks?" with "cerebellum", the LLM offers a detailed explanation of the underlying concept, such as clarifying the definition of neural networks. If the answer is correct, Mini-Xiaowei responds with positive feedback through facial expressions, further enhancing the learning motivation and engagement.

To enhance the learning motivation, the game incorporates two incentive mechanisms: "affinity points" and "exploration points". Students earn points by completing tasks such as watching instructional videos, interacting with Mini-Xiaowei, and answering questions correctly. The points with each student's behavioral sequences are compiled into a report that is automatically uploaded to the teacher interface, facilitating subsequent ranking and personalized guidance.

From the technical perspective, the system stores key contextual information as game variables to maintain dialogue coherence. Through this design, this study achieves a deep integration of DGBL and LLM-based scaffolding, contributing to advancing interactive and personalized AI education to a higher level.

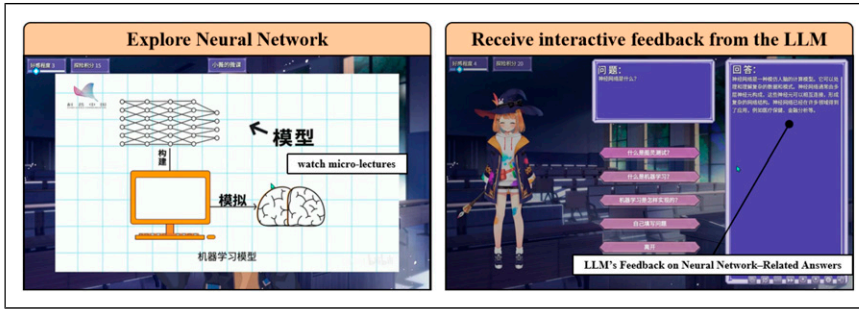


Figure 3. Explore Neural Networks and Receive Feedback From the LLM

LLM-Based Scaffolding Mechanism

The system is learner-centered and leverages LLM-based scaffolding to improve learning outcomes. It employs a locally deployed Ollama LLM service (developed in Go) that is integrated with the game engine via a TypeScript API. This integration provides intelligent support for various interface elements, including dialogue windows, scene interactions, and virtual characters. The LLM used is the Qwen model, which enables rapid responses and the generation of concise, contextually relevant guidance. During gameplay, it offers real-time Q&A, timely prompts, key information storage, and clue delivery (see Figure 4).

Throughout the exploration process, students can freely navigate content, consult resources, complete tasks, and take quizzes, while also interacting with intelligent virtual avatar "Mini-Xiaowei" to receive personalized feedback. Mini-Xiaowei will dynamically adjust its facial expressions and vocal tone in response to each learner's interactions to sustain engagement (see Figure 5). When incorrect responses are given, Mini-Xiao Wei will provide a detailed error analysis; when the answer is correct, it offers extended insights and positive reinforcement. Affinity points and exploration points with behavioral logs are recorded in the system's backend, offering teachers a visualized record of each student's trajectory.

The game incorporates leveled challenges, character dialogues, and multiple adaptive strategies to provide personalized support. These elements allow each student to engage in inquiry, critical thinking, and deep knowledge construction within an enjoyable and immersive learning environment.

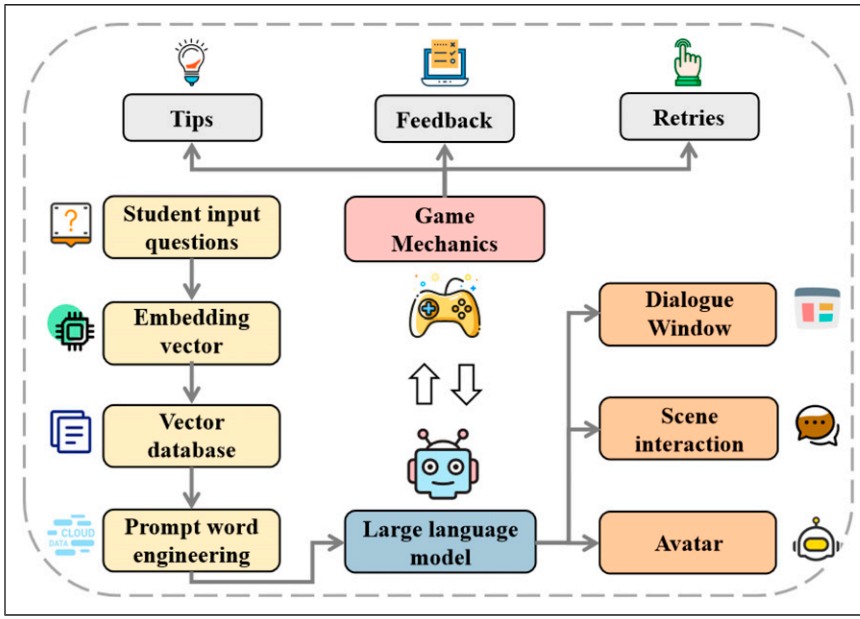


Figure 4. Illustration of LLM-Based Scaffolding Mechanism

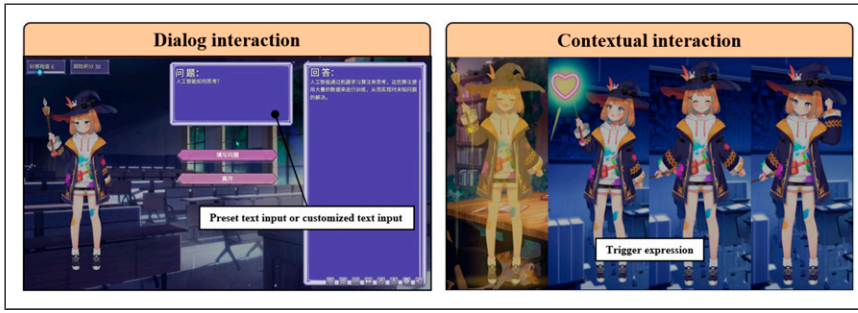


Figure 5. The Personalized Feedback and Facial Expressions of Mini-Xiaowei

Method

Participants

A quasi-experimental design was selected to preserve the integrity of existing class structures and align with the practicalities of conducting research within a school environment. A random sampling procedure was used to select two fifth-grade classes from an elementary school in eastern China, with a total of 80 students aged 12 to 13 participating in the experiment. All participants had at least three years of education in information technology and were proficient with DGBL tools. Based on their class groupings, students were divided into an experimental group and a control group (40 students each, with an equal gender distribution). The experimental group used LLM-DGBL, while the control group used TS-DGBL. Both groups were taught by the same teaching team to ensure consistency in teaching methods and pace. The experimental course was an elementary AI literacy course covering topics such as "What is AI?", "Thinking Like a Human," and "Understanding Human Language".

The key independent variable was the type of scaffolding provided. The experimental group received dynamic, personalized feedback through an LLM-based system (LLM-DGBL), whereas the control group received static, predefined hints from a fixed question bank (TS-DGBL). The full system design is detailed in the "Description of LLM-based scaffolding" section, and the experimental procedure is visualized in Figure 6.

Measuring Instruments

Both the pre- and post-tests used to assess academic achievement in this study were iteratively reviewed and revised through multiple rounds of expert consultation and feedback from course instructors to ensure clarity and comprehensibility. The pre-test assessed baseline knowledge across modules such as "What is AI?", "Thinking Like a Human", and "Understanding Human Language." The post-test was designed to compare learning achievement between the two groups under different instructional conditions. Each assessment comprised 20 illustrated multiple-choice items, with a total

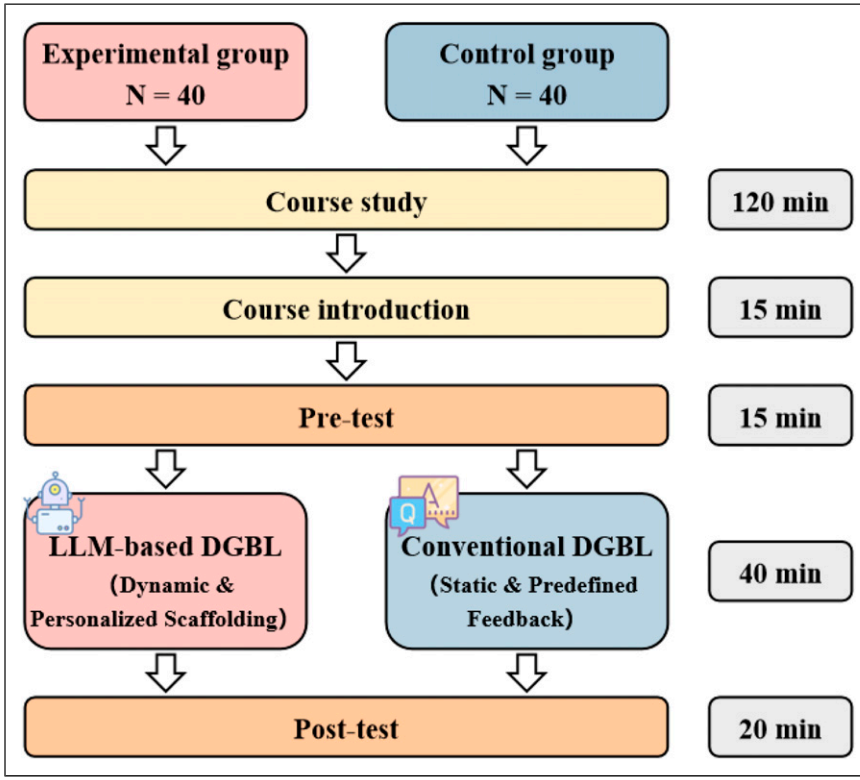


Figure 6. Diagram of Experiment Design

score of 100. Both the pre- and post-tests achieved a KR-20 reliability coefficient of 0.53, indicating an acceptable level of reliability, as values above 0.50 are generally considered adequate (Saidi & Siew, 2019).

The flow experience and cognitive load questionnaires were adapted from established scales using a consistent three-step process to ensure their validity and reliability. First, the original English items were translated into Chinese and subsequently back-translated by bilingual experts to ensure conceptual equivalence. Second, the wording of several items was contextualized to fit the AI learning game environment (e.g., replacing the generic term "activity" with "the learning game"). Finally, the content validity of both adapted questionnaires was established through a review by two educational technology experts and one course instructor, who verified the clarity, relevance, and appropriateness of the items for fifth-grade students. This rigorous adaptation process was particularly important given the challenges of measuring subjective states in young learners, such as their developing metacognitive awareness. The expert review ensured that the language was concrete and the concepts were understandable, thereby making the instruments suitable for this specific age group.

The flow experience questionnaire used in this study was adapted from Pearce et al. (2005), which was six-dimensional scale and consisted of 18 items rated on a five-point Likert scale. Sample items included: "I was deeply absorbed in the learning activity", "I found the learning activity highly engaging", and "I had to exert considerable effort to stay focused during the learning activity". These items assessed dimensions of perceived control, perceived interest, and perceived enjoyment. The questionnaire was confirmed with a Cronbach's alpha of 0.82, exceeding the commonly accepted threshold of 0.70, indicating a high level of internal consistency (Hair et al., 2012).

The cognitive load questionnaire was adapted from the classical scales developed by Paas (1992) and Sweller et al. (1998), with modifications based on Hwang et al. (2013). It included eight items, five assessed mental load and three assessed mental effort. All items were rated on a five-point Likert scale, with higher scores indicating greater cognitive load. For example, "Did the content of this learning activity present difficulty for you?" assessed mental load, while "Was the instructional guidance of this learning activity difficult to understand?" assessed mental effort. The questionnaire demonstrated strong internal consistency, with a Cronbach's alpha of 0.79, exceeding the commonly accepted threshold of 0.70 for acceptable reliability (Hair et al., 2012).

The behavioral coding scheme of LLM-based scaffolding. To analyze the learning behaviors within the digital game environment, this study builds on the coding frameworks of Crompton et al. (2018) and Liang et al. (2024), while incorporating the operational logic and task design of the game. First, the video-watching behavior was refined by distinguishing between normal and accelerated speed to prevent a decline in learning outcomes, as prior research indicated that students tend to skip content to access information more quickly when watching videos (Xu et al., 2024). Second, interactions with LLM were categorized into fixed and free talks. Fixed talks followed a predefined set of questions and response pathways to ensure that core instructional points were comprehensively covered, whereas free talks enabled students to raise questions based on their individual needs, providing personalized learning and deeper engagement.

Based on these considerations, a behavior coding scheme was developed (Table 1). To ensure both scientific rigor and practical applicability, the research team collaborated with two experienced teachers to systematically refine the operational definitions, coding criteria, and annotated examples for each behavioral indicator. During implementation, string variables were embedded within the game to capture and record all student interactions in real time. These data were then compiled for centralized storage. This approach not only improved the completeness and accuracy of data recording but also provided a robust foundation for subsequent quantitative analysis and behavioral pattern mining. Recognizing that the learning behaviors of elementary students can be more exploratory and less linear than those of older learners, our coding scheme was designed to capture this nuance. For example, distinguishing between "fixed talk" and "free talk" allowed us to analyze both structured inquiry and spontaneous exploration, providing a developmentally sensitive framework for interpreting their learning processes.

Table 1. Coding Scheme of LLM-Based Scaffolding Behavior

Code	Behavior	Description
I	View micro-lecture	Click to watch the instructional micro-lecture
D	Accelerated viewing	Click the speed button to watch the micro-lecture at accelerated speed
O	Normal-speed viewing	Click the normal-speed button to watch the micro-lecture at normal speed
Q	Start a task	Click the "Explore Map Task" button to begin the game task
H	Summon Mini-Xiaowei	Summon the intelligent virtual avatar "Mini-Xiaowei"
ST	Talk with Mini-Xiaowei (fixed)	Summon Mini-Xiaowei in a fixed talk
FT	Talk with Mini-Xiaowei (free)	Summon Mini-Xiaowei in a free talk
QT	Quit conversation	Open mini- Xiaowei's chat window and select "Leave"
S	Start answering questions	Start answering the questions
W	Wrong answer	Submit an incorrect answer
C	Correct answer	Submit a correct answer
M	Read knowledge card	Click to read the acquired knowledge card
L	Read task clue	Click to read the content of the hint box

Intervention

This study utilized an "Elementary General AI Course" as the instructional context to examine the effects of LLM-DGBL on learning outcomes (Figure 6). The experiment was conducted in a computer lab equipped with 50 computers of the same configuration. Prior to the formal experiment, the installation and configuration of the educational game were completed to ensure stable system operation.

Before the DGBL session, students received two weeks of instruction on the elementary general AI course (120 minutes). After the course, the teacher spent 15 minutes introducing the objective and procedure of the experiment, after which students spent 15 minutes completing the academic pre-test to assess their baseline knowledge level.

During the 40-min DGBL session, students engaged in their assigned conditions (LLM-DGBL or TS-DGBL). As noted previously, all other elements of the game—including scenarios, content, and task design—were identical for both groups.

Following the game session, students spent 20 minutes completing the post-test and the questionnaire on flow experience and cognitive load. By comparing test scores and questionnaire responses across the two learning conditions, the effectiveness of LLM-based scaffolding in enhancing learning outcomes was evaluated.

Ethical Considerations and Risk Mitigation

In alignment with the RAISE framework and acknowledging the ethical sensitivities inherent in conducting AI-mediated research with elementary school students

(Allison, 2025), we identified potential risks and embedded corresponding safeguards into our study design. Our approach to mitigating risks across pedagogical, equity, and data governance themes is detailed in Appendix. The implementation of these safeguards was integral to maintaining the ethical and scholarly integrity of our research. Proactively addressing these socio-technical dimensions was crucial for creating a learning environment that was effective, engaging, safe, equitable, and respectful of learner rights.

Data Analysis

To address the research questions, statistical analyses were performed using SPSS 26.0, with GSEQ 5.1 used for lag sequential analysis. An alpha level of .05 was set for all significance tests.

First, the Shapiro-Wilk test was used to check the normality of the data. As the data did not meet the assumption of normality, a series of non-parametric tests were employed. To assess the intervention's primary effects, Mann-Whitney *U* tests were conducted to compare differences between the two groups. To evaluate the practical significance of these differences, we calculated the Rank-Biserial Correlation (*r*) as a measure of effect size. To make the tables clearer and focus the reader on the magnitude of the effect, we report its absolute value, |r|, in the tables. The direction of each comparison (i.e., which group scored higher or lower) is explicitly described in the narrative of the results section. The conventions for interpreting the magnitude of the effect size are: .1 (small effect), .3 (medium effect), and .5 (large effect). The scaffolding condition (LLM-DGBL vs. TS-DGBL) served as the independent variable, and the main learning outcomes (learning achievement, flow experience, and cognitive load) were the dependent variables.

To analyze students' behavioral patterns, lag sequential analysis (LSA) was conducted on the real-time interaction logs. Behavioral sequences with Z-scores exceeding 1.96 were identified as statistically significant transitions. Finally, to examine the interaction behaviors within the experimental group, following a common procedure for subgroup analysis (Sharma et al., 2025), we stratified students into high- and low-cognitive load subgroups using a median split. A Mann-Whitney *U* test was then used to compare the interaction frequency between these two subgroups. For all significant results, corresponding effect sizes were calculated and reported.

Results

The results are presented in four main sections. First, we compare the learning achievement, flow experience, and cognitive load between the experimental and control groups. Second, we present a lag sequential analysis of learning behaviors to reveal distinct interaction patterns. Finally, we examine how students with different cognitive load levels in the experimental group utilized the LLM-based scaffolding.

Table 2. The Mann-Whitney *U* Test Results of the Two Groups' Learning Achievement

Learning achievement	Group	*N*	Mean	Mean rank	*U*	*Z*	*p*	*r*
Pre-test	EG	40	54.88	44.30	648.00	−1.463	.139	.164
	CG	40	51.50	36.70				
Post-test	EG	40	72.88	47.39	524.50	−2.651	.008**	.296
	CG	40	61.75	33.61				

EG = Experimental Group; CG = Control Group. **$p < .01$. Effect sizes (r) are interpreted using Cohen (1988) 's conventions: 0.1 = small, 0.3 = medium, and 0.5 = large.

Learning Achievement

The LLM-based scaffolding significantly enhanced students' learning achievement. A Mann-Whitney *U* test revealed no significant difference in pre-test scores between the experimental group (EG) and the control group (CG), indicating comparable baseline knowledge ($U = 648.00$, $Z = -1.463$, $p = .139$, $r = .164$). However, after the intervention, the experimental group's post-test scores were significantly higher than those of the control group ($U = 524.50$, $Z = -2.651$, $p = .008 < 0.01$). This difference represented a medium effect size ($r = .296$), suggesting that the scaffolding had a practically significant impact on improving learning outcomes in the DGBL environment (see Table 2).

Flow Experience

The LLM-based scaffolding did not significantly alter students' flow experience compared to the control condition. A Mann-Whitney *U* test showed no significant difference between the groups ($U = 650.50$, $Z = 1.439$, $p = .148 > 0.05$), with a small effect size ($r = .161$) (see Table 3). The mean scores for the experimental group ($M = 4.17$) and the control group ($M = 3.97$) were both high, suggesting that the interactive and engaging nature of the gamified learning environment itself was effective in fostering a high level of immersion for all participants. This result did not replicate the findings of (Wang et al., 2025), indicating that the impact of LLMs on flow experience may be context-dependent.

Cognitive Load

The LLM-based scaffolding effectively reduced students' cognitive load. The results of a Mann-Whitney *U* test showed a statistically significant difference between the two groups, with the experimental group (Mean Rank = 34.16) reporting a significantly lower cognitive load than the control group (Mean Rank = 46.84; $U = 546.40$, $p = .014 < 0.05$). The effect size approached a medium level ($r = .273$), confirming the practical significance of the scaffolding in alleviating cognitive burden during the DGBL process (see Table 4). This aligns with prior

Table 3. The Mann-Whitney *U* Test Results of the Two Groups' Flow Experience

Group	*N*	Mean	Mean rank	*U*	*Z*	*p*	*r*
EG	40	4.17	44.24	650.50	1.439	.148	.161
CG	40	3.97	36.76				

EG = Experimental Group; CG = Control Group. Effect sizes (*r*) are interpreted using Cohen (1988) 's conventions: .1 = small, .3 = medium, and .5 = large.

research demonstrating the positive role of scaffolding in digital learning environments (Faber et al., 2024).

Learning Behavior

Lag sequential analysis revealed distinct patterns of engagement between the two groups (see Tables 5 and 6, Figures 7 and 8). To analyze students' dynamic learning processes, their in-game actions were logged and converted into behavioral sequences. Lag sequential analysis was then used to identify statistically significant behavioral transitions ($Z > 1.96$). In total, 2,839 behaviors were recorded for the experimental group and 1,685 for the control group.

Experimental Group: Active and Iterative Help-Seeking. The defining characteristic of the experimental group was their frequent and purposeful interaction with the AI avatar, "Mini-Xiaowei." Significant behavioral chains showed students summoning the avatar after viewing lecture content (D → H, O → H), engaging in repeated dialogues (ST → ST, FT → FT), and seamlessly transitioning between conversation and other learning tasks (H → ST/FT, H → QT). This active help-seeking was integrated into a broader, cyclical learning pattern of "viewing content → seeking clarification → reading clues → answering questions."

Control Group: Linear and Direct Content Progression. In contrast, the control group's behavior was more linear and straightforward. Their dominant patterns involved a direct path from starting a task to viewing a lecture and then reading a clue (Q → I → D/O → L). Their engagement was primarily focused on consuming the provided content and then proceeding to answer questions, without the iterative, exploratory loops observed in the experimental group.

Table 4. The Mann-Whitney *U* Test Results of the Two Groups' Cognitive Load

Group	*N*	Mean	Mean rank	*U*	*Z*	*p*	*r*
EG	40	1.82	34.16	546.40	2.439	.014*	.273
CG	40	2.10	46.84				

EG = Experimental Group; CG = Control Group. *$p < .05$. Effect sizes (*r*) are interpreted using Cohen (1988) 's conventions: .1 = small, .3 = medium, and .5 = large.

Table 5. Results of Residuals Transition in Experimental Group

	I	D	O	Q	H	ST	FT	QT	S	W	C	M	L
I	−3.06	43.30*	27.15*	−2.82	−4.23	−5.87	−3.25	−4.41	−6.09	−4.07	−4.11	−0.25	−3.29
D	−2.56	−2.30	−1.44	−2.36	33.64*	−4.91	−2.72	−3.69	−5.10	−3.41	−3.44	−0.21	−2.75
O	−1.61	−1.44	−0.90	−1.48	21.09*	−3.08	−1.71	−2.31	−3.20	−2.14	−2.16	−0.13	−1.72
Q	47.31*	−2.36	−1.48	−2.43	−3.64	−5.04	−2.80	−3.79	−5.24	−3.50	−3.53	4.62*	−2.82
H	−4.11	−3.69	−2.31	−3.79	−5.68	7.95*	11.08*	20.30*	−8.18	−5.47	−5.52	−0.34	−4.41
ST	−5.47	−4.91	−3.08	−5.04	−7.55	37.13*	0.35	5.39*	−10.88	−7.27	−7.34	−0.45	−5.87
FT	−3.03	−2.72	−1.71	−2.80	−4.19	−2.95	21.99*	15.15*	−6.03	−4.03	−4.07	−0.25	−3.25
QT	1.61	−3.69	−2.31	6.67*	−5.68	−7.88	−4.37	−5.92	1.95	−5.47	−5.52	−0.34	38.57*
S	−5.68	−5.10	−3.20	−5.24	−7.84	−10.88	−6.03	−8.18	−11.30	34.59*	34.91*	−0.47	−6.09
W	−3.51	−3.15	−1.97	6.58*	−1.14	−6.72	−3.73	−5.05	23.84*	−4.66	−4.71	−0.29	−3.76
C	−3.76	−3.38	−2.12	15.72*	−0.76	−7.20	−4.00	−5.42	20.53*	−5.00	−5.05	−0.31	−4.04
M	4.26*	−0.21	−0.13	−0.22	−0.32	−0.45	−0.25	−0.34	−0.47	−0.31	−0.32	−0.02	−0.25
L	−3.06	−2.75	−1.72	−2.92	12.72*	−5.87	−3.25	−4.41	14.95*	−4.07	−4.11	−0.25	−3.29

*$Z > 1.96$; $P < 0.05$.

Table 6. Results of Residuals Transition in Control Group

	I	D	O	Q	S	W	C	M	L
I	0.00	35.80*	20.51*	−3.67	−10.74	−5.21	−6.05	0.91	−4.36
D	−3.18	0.00	−1.70	−3.15	−9.21	−4.46	−5.18	−0.60	35.61*
O	−1.72	−1.70	0.00	−1.69	−4.96	−2.40	−2.79	−0.32	20.22*
Q	41.47*	−3.15	−1.69	0.00	−9.16	−4.44	−5.16	−0.59	−3.71
S	−9.27	−9.21	−4.96	−9.16	0.00	14.57*	15.95*	1.62	−10.87
W	−4.27	−4.24	−2.28	8.75*	7.96*	0.00	−6.95	−0.80	−5.01
C	−5.01	−4.97	−2.68	14.93*	5.94*	−7.02	0.00	−0.94	−5.87
M	−0.60	1.23	−0.32	−0.59	−1.74	3.31*	0.26	0.00	−0.71
L	−3.76	−3.73	−2.01	−3.71	13.12*	−5.27	−6.12	−0.71	0.00

*$Z > 1.96$; $P < .05$.

Shared Foundational Behaviors. Both groups shared several foundational learning behaviors. For example, students in both conditions tended to view micro-lectures after starting a task (Q → I), adjust video playback speed (I → D/O), and begin answering questions after reading a clue (L → S). They also followed similar patterns after answering a question, either continuing to answer or moving to a new task (e.g., C → S, W → S, C → Q).

The key distinction was the experimental group's proactive use of the LLM-based scaffolding to create feedback loops, which fostered deeper cognitive engagement. This pattern of cycling through instruction, interaction, and application demonstrates the significant impact of the scaffolding in transforming a standard gamified learning process into a more dynamic and supportive experience.

Scaffolding Usage by Students with Different Cognitive Loads

To further understand the utility of the scaffolding, we analyzed its usage frequency between students with high and low cognitive loads within the experimental group.

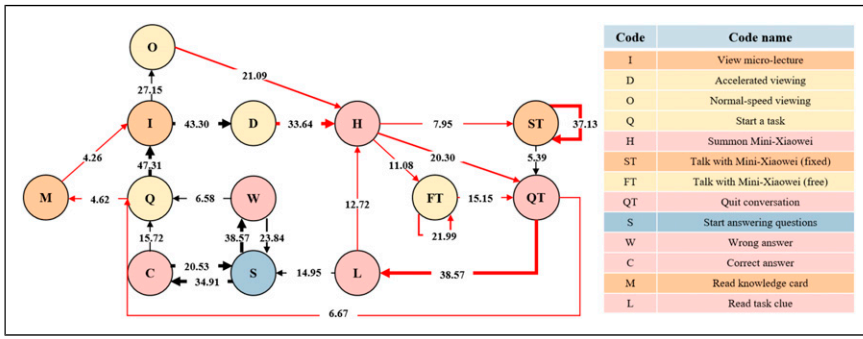


Figure 7. Behavioral Transition Diagram of Experimental Group

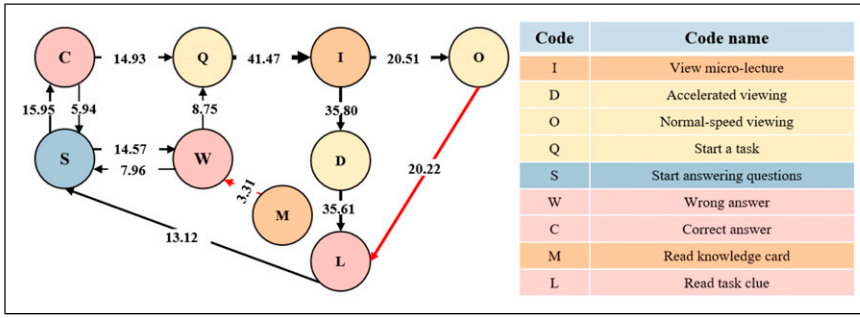


Figure 8. Behavioral Transition Diagram of Control Group

Students with high cognitive load interacted with the scaffolding significantly more often. As shown in Table 7 and Figure 9, the high-load subgroup ($M = 23.44$) used the scaffolding nearly three times as frequently as the low-load subgroup ($M = 8.45$). A Mann-Whitney U test confirmed this difference was statistically significant ($U = 73.0$, $p = .0007 < .001$) and represented a large effect size ($r = .537$). This finding strongly suggests that the LLM-based scaffolding served as a critical support tool for students facing greater learning challenges. Their higher interaction frequency reflects a clear need for assistance, providing empirical evidence for optimizing scaffolding designs based on learners' cognitive states.

Discussion

This study addressed a critical empirical gap by investigating the effects of LLM-based scaffolding within a DGBL environment. The findings provide crucial, comparative evidence that clarifies its advantages over traditional static scaffolding. Primarily, the LLM-based scaffolding significantly improved students' learning achievement and effectively reduced their cognitive load compared to traditional static scaffolding. Behavioral analysis further revealed that students in the experimental group adopted more complex and effective self-regulated learning strategies. Interestingly, no significant difference was observed in flow experience, as both groups reported high levels

Table 7. The Mann-Whitney U Test Results for Interaction Number Between the Two Subgroups

Group	*N*	Mean	Median	*SD*	Mean rank	*U*	Z	*p*	*r*
High cognitive load group	18	23.44	21.00	16.73	27.44	73.0	−3.398	.0007***	.537
Low cognitive load group	22	8.45	7.00	8.46	14.82				

***$p < .001$. Effect sizes (r) are interpreted using Cohen (1988) 's conventions: .1 = small, .3 = medium, and .5 = large.

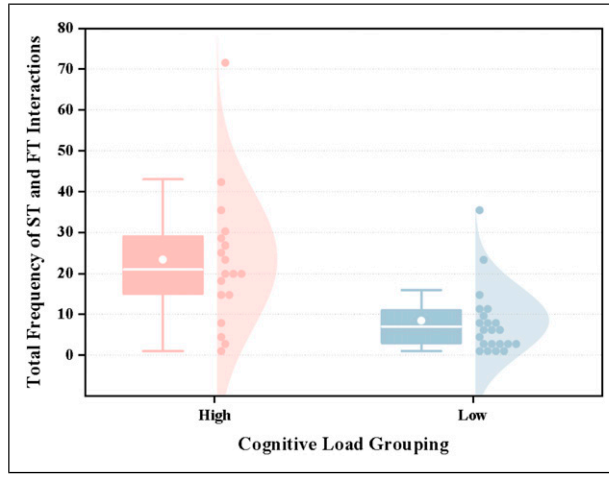


Figure 9. Comparison of Interactions Between High and Low Cognitive Load Student

of immersion, suggesting the underlying game environment was highly engaging for all participants. We will now discuss the implications of these findings in relation to existing literature.

The primary finding—that LLM-based scaffolding significantly enhances learning achievement—underscores its value in emerging disciplinary fields like AI education. This result aligns with research highlighting the power of personalized, immediate feedback in refining students' conceptual frameworks and correcting cognitive biases (Powell et al., 2024; Zhong, 2022). By providing rich, interactive feedback channels, the LLM not only made learning materials more accessible but also deepened students' engagement with the instructional content, thereby boosting learning efficiency (Goslen et al., 2024; Wang et al., 2025). This confirms the suitability of LLM-based approaches for delivering complex knowledge to young learners.

Regarding cognitive load, our results demonstrated that the experimental group experienced a significantly lower cognitive burden. In the control group, the absence of dynamic support likely increased task difficulty, leading to extraneous cognitive load. Conversely, the LLM-based scaffolding streamlined the learning process by offering on-demand clarification, thus allowing students to allocate their cognitive resources more effectively to the learning task itself (Leppink et al., 2013). This finding provides practical guidance for designing intelligent learning systems that are not only effective but also cognitively efficient, preventing the frustration and reduced performance associated with cognitive overload (Berends & van Lieshout, 2009). Thus, our findings provide an empirical contribution to Cognitive Load Theory by demonstrating how dynamic, LLM-based scaffolding can serve as an effective mechanism for managing extraneous load, thereby allowing learners to allocate more cognitive resources toward knowledge construction.

Furthermore, the behavioral analysis revealed a stark contrast in learning patterns. The control group followed a linear, passive cycle of "view → read → answer,"

whereas the experimental group demonstrated more sophisticated self-regulated learning behaviors. They actively managed their learning by adjusting playback speeds, seeking clarification from the AI avatar, and iteratively refining their problem-solving strategies. This frequent interaction with the AI reflects a shift from relying on external scaffolding to developing internal metacognitive monitoring (Podolefsky et al., 2014), a crucial skill for lifelong learning.

A deeper analysis within the experimental group provides a nuanced understanding of the scaffolding's role. We found that students with higher cognitive load interacted significantly more with the AI avatar. This does not suggest the scaffolding failed; rather, it indicates that the scaffolding was functioning precisely as intended. The increased interaction frequency is not a sign of a reduced challenge, but rather a reflection of students' active use of the tool to cope with a perceived high level of difficulty (Feldon, 2007). When task demands exceeded their cognitive capacity, these students strategically sought external support to manage their intrinsic cognitive load, avoid unproductive trial-and-error, and facilitate knowledge reconstruction (Jiang & Yu, 2025; Yan et al., 2025). This finding validates the scaffolding's role as an essential support mechanism for learners who need it most.

Beyond these findings, our study offers significant practical implications specifically for elementary education. Our results suggest that LLM-DGBL is particularly well-suited for this age group because it aligns with their developmental characteristics. Elementary learners benefit from the immediate, interactive feedback that makes complex topics more accessible and lowers the cognitive barrier to entry (Borghi et al., 2017; Shute, 2008). The dynamic, on-demand support of the LLM helps manage the cognitive load of complex AI concepts, making them more accessible to young minds. From a pedagogical standpoint, these findings have direct implications for elementary-level curriculum design, teacher facilitation, and learner autonomy. For curriculum design, our approach enables more flexible, inquiry-based models where LLM-based scaffolding can support differentiated learning paths. For teacher facilitation, the educator's role shifts from instructor to mentor, who must guide students to critically engage withh and verify AI-generated content, balance structured tasks with open-ended exploration, and prepare to intervene if the AI provides incorrect answers, reinforcing their indispensable role in the classroom. Finally, by providing a non-judgmental, "safe" space to ask questions, the scaffolding fosters greater learner autonomy, empowering students to take ownership of their exploration of AI.

However, realistically integrating such scaffolding into real-world elementary classrooms requires careful consideration of several practical factors. Challenges may include unequal access to technology among schools, the need for robust teacher training to help educators transition into a facilitator role, and ensuring the system remains a supportive tool rather than a crutch that hinders genuine problem-solving skills. To address these, we propose several instructional strategies. First, feasibility can be enhanced by using locally deployed, open-source LLMs (as was done in this study) to reduce costs and ensure data privacy. Second, teachers should be trained not only on the technical aspects but also on pedagogical strategies, such as prompting students to question and verify AI-generated information. Finally, the system can be integrated into

classroom practice in various phases: (1) as a pre-task tool to activate prior knowledge; (2) as an on-demand "digital teaching assistant" during tasks, freeing up the human teacher for targeted support; and (3) as a post-task diagnostic tool, where teachers review interaction logs to identify common misconceptions. This phased integration ensures the technology serves as a practical enhancement to, rather than a disruption of, existing classroom workflows.

Limitations and Future Research

Despite these promising findings, several limitations of this study point to opportunities for future research. One limitation lies in the study's scope and context. The intervention was implemented in a relatively short session within a single school, which may constrain the generalizability of the findings across different learning durations, diverse student populations, and educational settings. Additionally, while our behavioral analysis offered valuable insights, incorporating qualitative data, as suggested for future research, would further deepen the understanding of students' learning processes. In addition, the learning achievement assessments used in this study could be further refined to enhance measurement precision and reliability. Methodologically, our design compared a dynamic LLM-based system with a static environment, which may not fully disentangle the specific effects of generative AI from those of general interactivity. Finally, the behavioral analyses, relying mainly on quantitative log data, may not fully capture the complexity of learners' cognitive and motivational processes.

Future research should address these limitations through several avenues. Methodologically, there is a need for greater rigor, including the development and validation of more reliable measurement instruments through processes like item quality review and discrimination analysis, as well as the use of pilot testing for both interventions and assessments. Building on this foundation, longitudinal studies across diverse school settings are essential to evaluate the long-term impact of LLM-DGBL. To better isolate the unique contributions of the LLM, subsequent experimental designs could compare LLM-based scaffoldings to other advanced, non-generative interactive support systems. Furthermore, to deepen the interpretation of learning behaviors, future work should triangulate quantitative log data with qualitative methods, such as think-aloud protocols or stimulated recall interviews. Finally, the scope of inquiry should be broadened to explore the scaffolding's impact on other crucial dimensions of AI literacy, such as creative and critical thinking. Given that AI outputs require careful verification, this line of inquiry should also investigate the indispensable role of teacher guidance in helping students critically assess, question, and verify AI-generated content, thereby fostering genuine digital literacy.

Conclusion

In conclusion, this study provides strong empirical evidence for the effectiveness of integrating LLM-based scaffolding within a DGBL environment for elementary AI education. By demonstrating significant in learning achievement, reductions in

cognitive load, and the promotion of self-regulated learning behaviors, our findings highlight the transformative potential of generative AI in creating learning experiences for young learners that are not only personalized and supportive but also cognitively and developmentally appropriate. By addressing their specific learning needs—such as the need for concrete examples and immediate feedback—this study demonstrates a pathway to making the complex concepts of AI more accessible and manageable for elementary students.

From a practical standpoint, this research offers a feasible framework for educators and instructional designers looking to implement generative AI in the classroom. For optimal impact, we recommend that such systems be designed not as replacements for instruction, but as dynamic support tools that empower students to manage learning challenges proactively. The finding that high-load students are the most active users suggests that these tools are particularly valuable for differentiation and providing targeted support. Ultimately, this study serves as a critical step toward harnessing the power of LLMs to foster a new generation of learners who are not only AI-literate but are also more efficient and self-regulated in their approach to complex knowledge.

Acknowledgments

Many thanks for all the teachers' and students' work.

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Ethical Considerations

This research was conducted in accordance with the general ethical guidelines prevalent in the field of science education research. Our research protocol was tailored to benefit both students and teachers. Informed consent was obtained from teachers, students, and parents, and they were provided with the option to decline participation in any aspect of this research project. Furthermore, all participant identities have been anonymized to safeguard the privacy of both students and teachers.

Consent to Participate

Consent was obtained from all students included in the study. Data was reported anonymously in the study.

Author Contributions

Yulin Gong: conceptualization, methodology, data curation, formal analysis, writing – original draft, reviewing and editing, project administration. Minkai Wang: conceptualization, methodology, writing – original draft, reviewing and editing. Li He: conceptualization, reviewing and editing. Chengshu Xu: conceptualization, methodology, writing – reviewing and editing, funding acquisition. Yue Yu: conceptualization, methodology, data curation.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The datasets analyzed during the current study are available from the corresponding author upon reasonable request.

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Appendix

Ethics and Risk Matrix for the LLM-DGBL Study (Adapted from Allison (2025)).

Risk theme	Risk category	Observed or anticipated impact	Mitigation/Safeguards
Pedagogical integrity and learning impact	Hallucination/ Misinformation	Anticipated: The LLM-powered avatar "Mini-Xiaowei" could potentially generate inaccurate or misleading information about AI concepts, confusing young learners	Mitigation: 1. Utilized a locally deployed Ollama LLM service with the Qwen model, providing a controlled knowledge base 2. Designed "fixed talk" pathways to ensure core instructional concepts were delivered accurately and comprehensively 3. Teachers could monitor student progress via the backend interface to provide timely guidance and clarification
	Loss of learner agency	Anticipated: Students might develop an over-reliance on "Mini-Xiaowei" for answers, thereby reducing their effort in critical thinking or engaging with primary learning materials	Mitigation: 1. The DGBL design mandated the completion of core tasks, such as watching micro-lectures, which could not be skipped 2. The LLM-based scaffolding was positioned as a supportive, on-demand tool rather than the primary mode of instruction
	Emotional impact and wellbeing	Anticipated: Automated feedback, if perceived as negative, could potentially cause anxiety or discouragement among students	Mitigation: The avatar "Mini-Xiaowei" was designed to provide positive reinforcement, using encouraging facial expressions and tone of voice for correct answers to enhance motivation and engagement

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Risk theme	Risk category	Observed or anticipated impact	Mitigation/Safeguards
Equity, inclusion, and accessibility	Accessibility barriers	Anticipated: Students with diverse learning preferences or reading difficulties might struggle with a text-only interface	Mitigation: The system was designed for multi-modality. Textual outputs from the LLM could be converted into speech via the iFlytek API, allowing students to switch between auditory and visual modes based on their preference
Data governance and learner rights	Privacy and data security	Anticipated: Risks of unauthorized access or data leakage concerning student interaction logs, performance data, and behavioral sequences	Mitigation: 1. The entire system, including the LLM, was locally deployed, preventing student data from being transmitted to external third-party servers 2. This study was reviewed and approved by the China Institute for Rural Education Development at Northeast Normal University on May 27, 2024
	Informed consent and autonomy	Anticipated: Ensuring that young participants and their legal guardians fully understood the nature of the research and their rights	Mitigation: All students and their guardians were fully informed of the voluntary nature of their participation and their right to withdraw from the study at any time without penalty